UDEERN VARANASI

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EDUCATION

The University of Texas at Austin — Bachelor of Biomedical Engineering; Minor in Material Science Engineering — May 2028

Relevant Coursework: Numerical Methods, Engineering Statistics, Thermodynamics, Biomechanics, Statics, Biomaterials

PROFESSIONAL EXPERIENCE

The Stachowiak Lab - *Undergraduate Research Assistant* — Aug 2025 – Present

- Engineered a Python pipeline to analyze microscopy videos (3,000+ frames), reducing analysis time by 90% per batch using subprocess calls and generated BeanShell scripts to automate ImageJ/Fiji and Filopodyan processes.
- Built NumPy/Pandas workflows for feature extraction and automated CSV, image-overlay, and PowerPoint reporting; implemented timeouts, status tracking, CSV validation, and diagnostic error logging.
- Coded MATLAB statistical-analysis models to establish the accuracy of pipeline-generated measurements. No statistically significant differences were observed when compared to prior work in the field.
- Presenting results as a poster at the UT BME Research Symposium (Dec 15, 2026).

The University of Texas at Austin - *BME 303L Teaching Assistant* — Aug 2025 - Dec 2025

- Taught 30+ students data acquisition, instrumentation, and physiological modeling in laboratory sessions.
- Diagnosed lab setup and measurement errors to help students troubleshoot data-collection problems.
- Explained experimental results and troubleshooting methods to help students interpret measurement data

The Neptune Lab - *Undergraduate Research Assistant* — Jan 2025 - Aug 2025

- Processed and analyzed motion-capture and force-plate datasets from 20 participants across 80 walking trials.
- Quantified ground and joint reaction forces in Vicon for comparison across walking conditions.
- Compared forward- and backward-walking force profiles to characterize gait differences.

Other Research - *Independent Researcher* — Aug 2025 – Present

ECG Time-Series Classifier

- Built a Python/Keras deep learning classifier for four-class arrhythmia prediction from 2,000-sample ECG sequences.
- Implemented a four-layer 1D convolutional neural network (CNN) for multiscale feature extraction, using batch normalization, dropout, and pooling for regularization and dimensionality reduction.
- Trained the model with the Adam optimizer and evaluated classification performance on held-out validation data, achieving 73.60% validation accuracy and 79.83% training accuracy.

Wearable Saccadic Movement Tracker

- Developed embedded C++ firmware on an Arduino Nano microcontroller for infrared sensor data acquisition and eye-movement detection in a wearable tracker.
- Led algorithm development for a five-person team; built a MATLAB computer-vision pipeline using grayscale conversion, percentile-based thresholding, and morphological operations to isolate pupil regions.
- Automated geometric feature extraction and data export, computing pupil centroids, axis lengths, and orientation and producing annotated images, binary masks, and .mat files for downstream tracking analysis.

2D Heat Diffusion Solver

- Implemented a finite-difference solver for the transient 2D heat equation on a 21-by-21 grid, computing temperature updates with a five-point discrete Laplacian.
- Parameterized fixed-temperature boundary conditions through an input vector for different simulation configurations; used separate state matrices and a stability-constrained time step for numerical updates.
- Generated 3D surface plots and time-lapse visualizations to inspect temperature evolution throughout each simulation.

SKILLS

Skills: Python, C++, MATLAB, R, Machine Learning, 1D CNNs, Signal Processing, Feature Extraction, Model Training and Validation, Statistical Analysis, Numerical Methods, ImageJ, Excel, Vicon